

length strings. `s[start:end]` gives us the substring of string `s` starting at index `start` to the index `end`, but not including the `end`. All indexes are zero-based indexes.

- `>>> s = "david"`
- `>>> s[0:4]`
- `'davi'`

If either of the indexes is not passed to the slicing operation, it defaults them to `start` or the `end` of the string respectively.

- `>>> s[2:]`
- `'vid'`

Extracting a character at an index from the string can be done in a similar fashion with a slightly different format. `s[index]` extracts the character at position '`index`' in string `s`.

- `>>> s = 'goliath'`
- `>>> s[1] # zero-based indexing`
- `'o'`

String Methods

The string class provides multiple methods to perform analysis, transformation and manipulation operations. Let's take a look at a few of them.

- Strings can be transformed to an all upper or lower case using the method `upper()` and `lower()` respectively.

A more aggressive form of lower casing can be achieved using the `case-fold()` method which is capable of performing operations on a range of characters that lie outside the 26-character set of English alphabets. It attempts to normalize a vast range of Unicode Characters such that they are much more comparable.

- `>>> "ABC".lower()`
- `'abc'`
- `>>> "xyz".upper()`
- `'XYZ'`